

6. CONTROL & CO-ORDINATION

❖ INTRODUCTION

- Control and coordination is the process by which different activities of an organism are regulated and integrated so that it can respond properly to internal and external stimuli.
- Living organisms perform many activities at the same time. These activities must be controlled and coordinated to maintain balance and survival.
- All the living organisms respond and react to changes in the environment around them.
- The changes in the environment to which the organisms respond and react are called stimuli such as light, heat, cold, smell, touch etc.
- Both plants and animals respond to stimuli but in a different manner.

Control and Coordination in Animals

- It is brought about in all animals with the help of two main systems:
 - a) Nervous system
 - b) Endocrine system

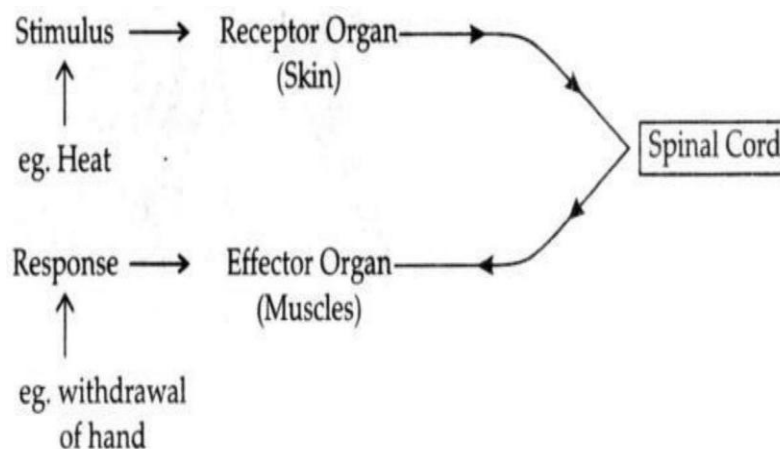
❖ Nervous System

- Control and coordination are provided by nervous and muscular tissues.
- Nervous tissue is made up of an organized network of nerve cells or neurons, and it is specialized for conduction of information via electrical impulses from one part of the body to another.
- Nervous system is the system of conducting tissues that receives the stimulus and transmits it to other parts of the body forming a network of nerves.
- The units which make up the nervous system are called **nerve cells** or **neurons**.
- The receptors pass the information to the brain through a type of nerve cells called **sensory neurons**.
- Motor neurons transmit the information from the brain to the effector organs, mainly muscles and glands.
- **Nerve Impulse:** It is the information in the form of chemical and electrical signals passing through neurons. These impulses are carried by dendrites towards the cell body.

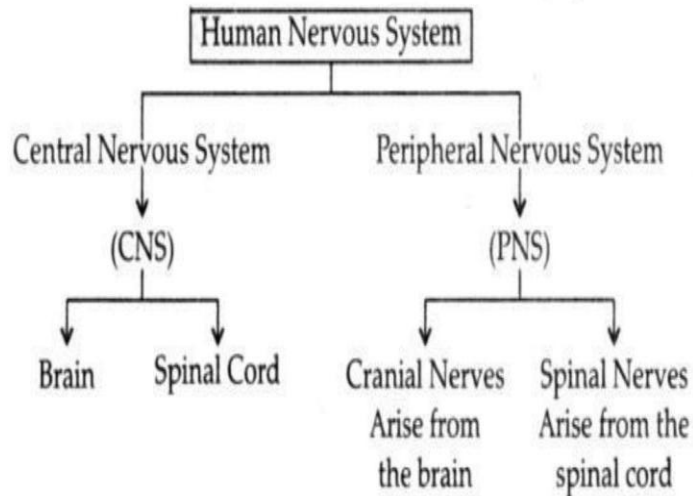
- **Neuromuscular Junction:** It is the point where a muscle fibre comes in contact with a motor neuron carrying nerve impulses from the central nervous system. The impulses travel from the neuron to the muscle fibres by means of neurotransmitter in the same way as the transmission of impulses across a synapse between two neurons.
- **Voluntary Action:** These are the actions which need thinking and are performed knowingly i.e. these are controlled by conscious thought. Example: Speaking to a friend, writing a letter etc.
- **Involuntary Action:** These are not under the control of the will of an individual and are automatic response to a stimulus which is not under the voluntary control of the brain. Example: Touching a hot plate unknowingly.

❖ Reflex Action

- Reflex action is quick, sudden and immediate response of the body to a stimulus. E.g., Knee jerk, withdrawal of hand on touching hot object.
- Reflex arc: The pathway through which nerve impulse pass during reflex action is called reflex arc.



- **Response:** Responses are of three main types:
 - (a) Voluntary:** Controlled by fore brain. E.g., talking, writing.
 - (b) Involuntary:** Controlled by mid and hind brain. E.g., heartbeat, vomiting, respiration.
 - (c) Reflex action:** Controlled by spinal cord. E.g., withdrawal of hand on touching a hot object.
- **Need of Reflex Actions:** In some situations such as touching a hot object, pinching etc. we need to act quickly, otherwise our body would be harmed. Here response is generated from spinal cord instead of brain.



❖ HUMAN BRAIN

- Brain is the main coordinating centre of the body. It has three major parts:
 - (a) Fore-brain
 - (b) Mid- brain
 - (c) Hind-brain

(a) Fore-brain: It is the most complex or specialized part of the brain. It consists of cerebrum.

Functions :

- (i) Thinking part of the brain.
- (ii) Control the voluntary actions.
- (iii) Store information (Memory).
- (iv) Receives sensory impulses from various parts of the body and integrate it.
- (v) Centre associated with hunger.

(b) Mid-brain: Controls involuntary actions such as controls visual reflexes, controls auditory reflexes.

(c) Hind-brain: It has three parts:

(i) Cerebellum: Controls posture and balance. Precision of voluntary actions e.g., picking pen.

(ii) Medulla: Controls involuntary actions e.g., blood pressure, salivation, vomiting.

(iii) Pons : Involuntary actions, regulation of respiration.

- Autonomic Nervous System means 'Self governing nervous system', consists of a pair of chain of nerves and ganglia found on both the sides of the vertebral column.

- Spinal cord is a cylindrical structure and a part of the central nervous system. It is made up of nerves which supply information to think.

❖ Coordination in plants

- Plant Movements: The movements of the individual plant parts or organs of a plant like shoot, root etc. are due to some external stimuli like light, force of gravity, chemical substance, water etc.
- Tropic Movement : It is the directional growth or movement of a plant organ in response to an external stimulus. Growth towards the stimulus is positive tropism and growth away from the stimulus is negative tropism.

❖ Hormones in animals

Hormones are the chemical substances which co-ordinate and control the activities of living organisms and also their growth.

s.No.	Gland	Hormones	Functions	Target Site
1.	Hypothalamus	(i) Releasing hormones (RH) (ii) Inhibiting hormones	Regulates secretion of pituitary hormones.	Pituitary gland
2.	Pituitary Gland	(i) Growth hormone (GH)	Controls growth-dwarfism & gigantism.	Most tissues
3.	Thyroid Gland	(i) Thyroxm (ii) Triiodothyronine (iii) Thyrocalcitonin (TCT)	Basal metabolic rate, RBC formation. Regulated Ca level.	Body tissues
4.	Adrenal Gland	(i) Adrenaline (ii) Nor adrenaline (iii) Corticoids	Increase alertness, pupillary dilation, piloerection, sweating and heartbeat .	Body tissues
5.	Pancreas	(i) Insulin	Regulates glucose homeostasis stimulates glycogenesis , controls carbohydrate metabolism	Tissues

		(ii) Glucagon	Maintains, glucose levels, stimulates gluconeogenesis. Release of sugar from liver.	
6.	Testis	(i) Testosterone (ii) Androgens	Develops male reproduction organs & accessory sexual characters. Influence male sexual behavior.	Male body tissues
7.	Ovary	(i) Estrogen	Develops female reproductive organs, accessory sexual characters & female secondary behavior.	Female body tissues
		(ii) Progesterone	Support pregnancy, stimulates milk secretion.	

Plant Hormones : Are chemical compounds which help to coordinate growth, development and responses to the environment.

Main plant hormones are:

- (a) Auxin: Synthesized at shoot tip, helps the cells to grow longer involved in phototropism
- (b) Gibberellin: Helps in the growth of the stem.
- (c) Cytokinins: Promotes cell division present in greater concentration in fruits and seeds.
- (d) Absciscic Acid: Inhibits growth, Cause wilting of leaves, Stress hormone.

Questions :

1. The growth of pollen tubes towards ovules is the example of:
 - (a) Hydrotropism
 - (b) Geotropism
 - (c) Chemotropism
 - (d) Phototropism

Ans: (c) chemotropism

2. How do the shoot and roots of a plant respond to the pull of earth's gravity?

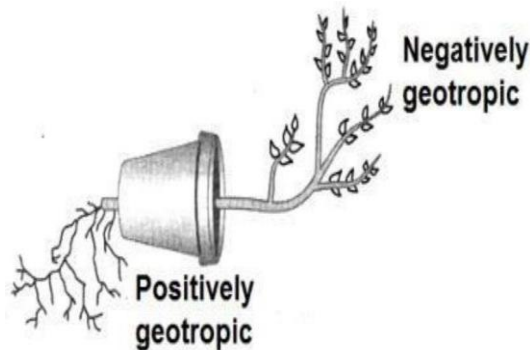
Ans: Roots grow downwards, towards gravity while shoot usually grow upwards and away from earth.

3. Explain the cause of shoots of the plant bending towards light.

Ans: Stems are positively phototropic and bend towards the direction of light. The movement is due to occurrence of more auxin on the darker side and lesser auxin on the illuminated side. Thus there is more growth on the darker side which causes the stem to bend towards light.

4. Define positive geotropism and negative geotropism. Give one example of each.
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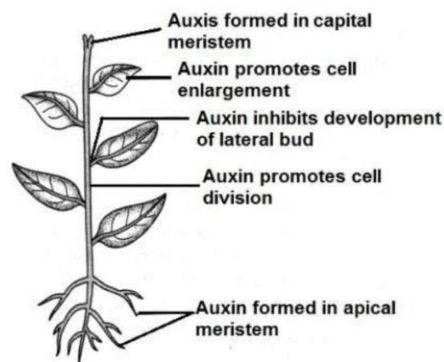
Ans: The movement of plant part in response to gravity is called geotropic movement and the phenomenon involved is called geotropism. When the tip of the stem grows away from the earth's gravitational forces, it is known as negative geotropism and when the root tips grow towards gravity, it is known as positive geotropism. E.g., Stem shows negative geotropism. Roots show positive geotropism.



5. How do auxins promotes the growth of a tendril around a support? Describe in brief.

Ans: Auxins produced in the shoot tip move downwards in the plant. These auxins cause cell elongation in the growing tissues. In the tendrils, auxins move away from the point of contact with the supporting object. More growth occurs on the side away from the support. As a result of unequal growth on the two sides, the tendril coils around the support.

6. Illustrate with the help of a diagram, the effect of auxins in different parts of a plant.



7. List in tabular form three differences in the movement of leaves of a Touch- me- not plant (the plant of Mimosa family) when touched and movement of a tendril towards a support.

S.No	Mimosa Leaves	Tendrils
(i)	Growth independent response	Growth dependent response
(ii)	It is a fast process	It is a slow process
(iii)	It is non - directional	It is directional

8. (a) Write the names and one function of each of any three growth hormones in plants.

(b) In the absence of muscle cells, how do plant cells show movement?

Ans: (a) Three growth hormones in plants are-

(i) Auxin: It is synthesized in the young tip of roots and shoots. It promotes elongation and division of cell and root formation.

(ii) Gibberellins: They help in the growth of the stem.

(iii) Cytokinins: They promote cell division and delay leaf ageing.

(b) The ability of a plant to detect change and respond to that change is termed as the sensitivity of the plant. Yet plants have no nervous system and no muscle tissue, they use electrical and chemical means to convey the information from one cell to another cell. The leaves of the sensitive plant (*Mimosa pudica*) folds up in response to touch. These leaf movements are independent of growth whereas, the directional movement of the shoot of a germinating seedling breaking through the soil is growth dependent.

9. Match the Column

Column - A		Column - B	
(i)	Reflex action	(a)	Thinking
(ii)	Stress hormone	(b)	Vomiting
(iii)	Cytokinin	(c)	Abscisic acid
(iv)	Hind brain	(d)	Cell division
(iv)	Fore brain	(e)	Sudden response

Ans: (i) - (e), (ii) - (c), (iii) - (d), (iv) - (b), (v) - (a).

10. Name the two components of peripheral nervous system.

Ans : Cranial nerves arising from the brain and the spinal nerves arising from the spinal cord.

11. How does our body maintain blood sugar level?

Ans : Timing and amount of hormone released are regulated by feedback mechanisms. If sugar level in blood rises, cells of pancreas detect and secrete more insulin which leads to the fall off sugar level.

12. (i) What is a reflex action?
(ii) Give example of involuntary action.

Ans: (i) Reflex action is defined as an unconscious and involuntary response of effectors to a stimulus.
(ii) Involuntary action: Beating of heart, salivation in the mouth on chewing of tasty food.

13. During exercise, the breathing rate is automatically enhanced. What is the reason behind it?

Ans: During exercise, the body demands more energy and oxygen. To fulfill these demands, the breathing rate is automatically enhanced, the heart beats faster, thus sending a greater volume of oxygenated blood to the muscles.

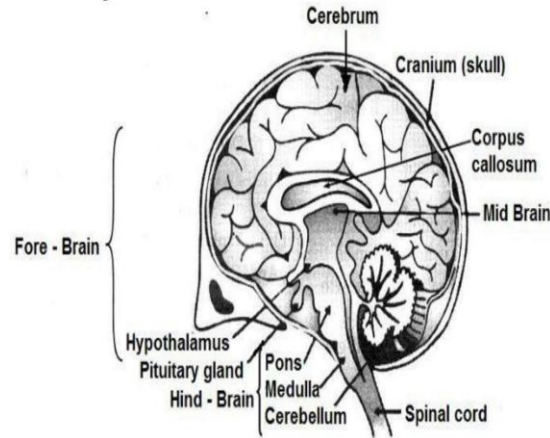
14. Draw a neat diagram of human brain and label on it the following parts:

- (i) Mid brain
- (ii) Pituitary gland
- (iii) Cerebellum
- (iv) Cerebrum

OR

- (i) Draw a well - labelled diagram of human brain.
- (ii) Which is the main thinking part of brain?

Ans: (i) Labelled diagram of Human brain



- (ii) Fore-brain.

15. Explain the feed back mechanism to regulate the action of the hormone s with the help of one suitable example.

Ans: Hormone s should be secreted in precise quantities. The timing and amount of hormone released are regulated by feedback mechanism. For example, if the sugar level in blood rises, they are detected by the cells of the pancreas which respond by producing more insulin. As the blood sugar level, falls, insulin secretion is reduced.

16. State three common features of respiratory organs of animals.

- Ans:** (i) All respiratory organs have large surface area to get enough oxygen.
(ii) They are thin-walled for easy diffusion of gases and substances.
(iii) They have rich supply of blood for transporting respiratory gases.

17. Write three main functions of the nervous system.

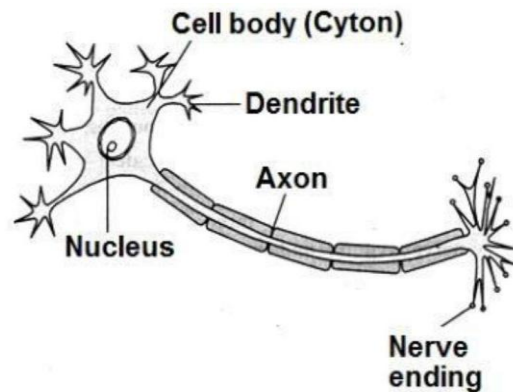
- Ans:** (i) Collects information from the surroundings.
(ii) Controls and co-ordinates the movement of muscles.
(iii) Helps to recognise smell, taste, vision, hearing etc. with the help of sense organs.

18. (a) Draw the structure of neuron and label cell body and axon.

(b) Name the part of neuron:

- (i) Where the information is acquired.
- (ii) Through which information travels as an electrical impulse.

Ans: (a)



- (b) (i) The information is acquired at the end of the dendrite tip of a nerve cell
(ii) The information travels as an electrical impulse from the dendrite to the cell body and then along the axon to its end.

19. 'Brain and Spinal cord are two vital organs of our body'. How is our body designed to protect them?

Ans: Brain and spinal cord are the parts of central nervous system. Brain is located inside a bony box, named as cranium and a fluid is present in this balloon, known as cerebrospinal fluid which is a shock absorber.

Spinal cord is placed in a vertebral column and back bone protects it.

20. (i) Define receptor and state their location in our body. Mention any two receptors present in our forebrain and their functions
(ii) How do nerve impulses travel in our body?

Ans: (i) Specialised tips of some nerve cells which receive information from the surroundings are receptor.

Location: Sense organs

Gustatory receptor: Detect taste

Olfactory receptor: Detect smell

(ii) The information acquired at the end of the dendritic tip of a nerve cell, sets off a chemical reaction that creates an electrical impulse, which travels to the cell body, then along the axon to its end, releases some chemicals, across the synapses.

21. (i) Write names of hormone s secreted by pituitary gland and adrenal gland. State their functions in the body.

(ii) Explain feedback mechanism for regulation of hormona l secretion with the help of one example.

Ans: (i) Pituitary gland secretes growth hormone. It regulates growth and development of the body. Adrenal gland secretes adrenaline hormone. It creates stress in our body.

(ii) The timing and amount of hormone s released by various glands is controlled by feedback mechanism which is inbuilt in our body, e.g. if sugar level in blood rises too high, it is detected by cells of pancreas which respond by producing and secreting more

insulin into blood. As blood sugar falls to certain level, the secretion of insulin is reduced automatically.

22. What happens at the synapse between two neurons?

Ans: At synapse the electrical impulse generated at dendrites of a neuron is passed on to dendrite of another neuron in the form of chemicals by axon ending of the first neuron.

23. Design an experiment to demonstrate hydrotropism.

Ans: Hydrotropism can be demonstrated with any germinating seedlings, which are allowed to grow on ground. The soil around the roots is separated by a polythene partition. The left side is kept moist and the right side dry. The radicle at first grows in a downward direction and after sometime the roots bend towards the moist soil. This is due to movement of the germinating roots to water.

24. Why is use of iodised salt advisable?

Ans: Iodine is necessary to make thyroxine hormone which regulates carbohydrates, protein and fat metabolism in the body so as to provide the best balance for growth.

25. Why are some patients of diabetes treated by giving injections of insulin?

Ans: Insulin regulates the sugar level in the blood. When this hormone is not synthesised in proper amount then the sugar level in the blood rises and the persons suffer from diabetes. This is the reason why diabetic patients are treated by giving injections of insulin.

Textbook Exercises

1. Which of the following is a plant hormone?

- (a) Insulin
- (b) Thyroxin
- (c) Estrogen
- (d) Cytokinin

Ans: (d) Cytokinin

2. The gap between two neurons is called a

- (a) dendrite
- (b) synapse
- (c) axon
- (d) impulse

Ans: (b) synapse

3. The brain is responsible for

- (a) thinking
- (b) regulating the heart beat
- (c) balancing the body
- (d) all of the above

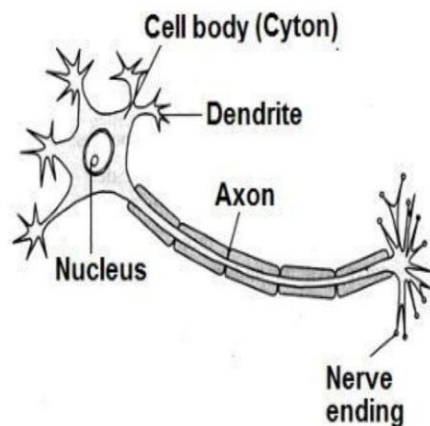
Ans: (d) all of the above

4. What is the function of receptors in our body? Think of situations where receptors do not work properly. What problems are likely to arise?

Ans: All information from environment is detected by receptors. When receptors do not work properly then information from environment cannot be detected and our body cannot respond accordingly.

5. Draw the structure of neuron and explain its function.

Ans: (a)



- (b) (i) The information is acquired at the end of the dendrite tip of a nerve cell
- (ii) The information travels as an electrical impulse from the dendrite to the cell body and then along the axon to its end.

6. How does phototropism occur in plants?

Ans: The chemical compounds released by stimulated cells for control and coordination in plants are called plant hormones or phytohormones.

Auxin: Controls growth

Gibberellins: Helps in growth of stem

Cytokinin: Promotes cell division

Absciscic acid: Inhibits growth, wilting of leaves.

7. Which signals will get disrupted in case of spinal cord injury?

Ans: In case of a spinal cord injury, signals for reflex action and involuntary action will get disrupted.

Reflex actions are monitored and controlled through the spinal cord of nervous system and not by the brain. In fact, nerves from all over the body meet in a bundle in the spinal cord on their way to the brain. In case of any injury to the spinal cord, the signals coming from the nerves as well as signals coming to the receptors will be disrupted.

8. How does chemical coordination occur in plants?

Ans: Chemical coordination occurs in plants by the help of plant hormones.

9. What is the need for a system of control and coordination in an organism?

Ans: It is necessary that various organs of the body of an organism work together in a proper manner to produce proper reaction to a stimulus.

10. How are involuntary actions and reflex actions different from each other?

Ans: Involuntary actions do not require thinking. It is controlled by the brain. For example, beating of heart. While on the other hand, the reflex action is rapid and spontaneous action in response to any stimulus. It doesn't involve the brain.

11. Compare and contrast nervous and hormonal mechanisms for control and coordination in animals.

Nervous system mechanism	Hormonal system mechanism
The information is conveyed in the form of electric impulse.	Information is conveyed in the form of chemical messengers.
The information is transmitted via axons and dendrites.	Information is transmitted through blood.
The flow of information is rapid and quick.	Flow of information is slow.
Its effects are short lived.	It has prolonged effects.

12. What is the difference between the manner in which movement takes place in a sensitive plant and the movement in our legs?

Ans: Sensitive plant, in response to touch, shows quick movement of leaves which fold up and droop. On the other hand, movement in legs is voluntary - action which is controlled by cerebellum part of hind brain.